Inferential Statistics is the branch of statistics that uses data from a sample to make conclusions, predictions, or generalizations about a larger population.

Main Objectives

- **Estimation**
  - Estimate **population parameters** based on Sample Data
  - Example: Estimating the average income of all citizens from a sample.
- **Test hypotheses (LEFT,RIGHT,TWO TAILED)**
  - Perform some test to jump to a conclusion based on sample data.
  - Example: Determining whether a new medicine is more effective than an existing one.
- **Regression Analysis (ML)**
  - Linear Regression
  - Logistic Regression

    Studies relationships between variables and makes predictions.

Important Topics

- **Central Limit Theorem**
- **Confidence Intervals**